

A Latency-Consistency-Aware Sub-Request Allocation Scheme for Boosting QLC SSD Read Performance

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Abstract—Abstract. Modern QLC SSDs leverage multi-level parallelism to enhance I/O performance by splitting requests into sub-requests dispatched on parallel units. Furthermore, the Independent Multi-Plane (IMP) read command is widely adopted to maximize plane-level parallelism by allowing simultaneous access to pages with different offsets. However, the existing parallel execution strategy substantially prolongs read response times due to two critical issues: 1) assembling sub-requests with different QLC page latencies into a single IMP command forces fast pages to stall for the slowest one, and 2) unbalanced queue lengths on parallel units create disparate waiting times for sub-requests from the same request. To address these issues, we propose a latency-consistency-aware sub-request allocation scheme called LCAS to significantly accelerate read performance by dynamically coordinating write allocation and read scheduling. During writes, LCAS allocates a request's sub-requests to QLC pages of the same type according to data hotness to reduce IMP command execution times. During reads, LCAS dynamically prioritizes near-complete requests into a single IMP command, effectively alleviating the impact of queue disparities. Evaluations against state-of-the-art schemes PACA and RRTS under eight real-world workloads demonstrate that LCAS optimizes the average read request completion time, sub-request wait time, and fast-page read access counts by averages of 27.8%, 29.71%, and 12.24%, respectively.

Index Terms—QLC Flash, SSD, Multi-plane operation, Data allocation

I. INTRODUCTION

NAND flash-based SSDs have become mainstream storage devices in modern systems due to their excellent read and write performance. To increase capacity and reduce costs, SSD manufacturers have developed multi-bit flash technology [1]. Currently, 4 bit/cell QLC flash memories have become the main storage medium for SSDs. Moreover, to further improve latency performance, SSDs are usually constructed with multiple channels, with each channel connecting to several NAND flash chips, and each chip containing multiple planes. This architecture allows SSDs to support multiple levels of parallelism, including the channel, chip, and plane levels [2], [3], [4].

To fully utilize these parallel resources, SSDs typically split a request into multiple page-sized sub-requests. These sub-requests are dispatched to chips for concurrent execution. Meanwhile, manufacturers developed multi-plane operations to further improve chip utilization [5], [6]. These commands

allow for the simultaneous execution of multiple operations (e.g., read, program, or erase) across all planes within a single chip. However, traditional multi-plane commands impose strict physical constraints, requiring all concurrent operations to be of the same type and their target addresses to have the exact same page offset [5], [7]. Consequently, their utilization rate for read operations is low, as it is highly difficult to gather concurrent read requests that share the exact same page offset. To overcome this restriction and maximize plane-level parallelism, the IMP read command was recently introduced [8]. This command allows for the concurrent reading of multiple pages with different offsets, enabling highly flexible utilization of plane-level parallelism.

Although modern SSDs utilize multi-level parallelism to improve performance, dividing a request introduces substantial synchronization waiting time. Specifically, SSDs dispatch the sub-requests of a request to multiple chips. To manage I/O execution, SSDs maintain an independent read queue for each plane and schedule sub-requests based on their arrival times. However, because queue lengths vary on different planes, sub-requests from the same request experience significantly different waiting times [9]. As a result, the entire request is delayed by its slowest constituent sub-request. Consequently, conventional scheduling schemes significantly increase the read response time.

Another critical inefficiency occurs when SSDs construct IMP read commands. To exploit plane-level parallelism, SSDs typically employ a FIFO policy, selecting pending sub-requests from each parallel unit's queue and groups them into a single IMP command. However, the FIFO policy ignores a fundamental characteristic of QLC NAND flash. Since each QLC cell contains four different bits, it is divided into four different page types (LSB, CLSB, CMSB, and MSB pages), which have significant differences read latencies [10]. Because an IMP command finishes only after completing the slowest read operation, the completed fast page reads are forced to stall and wait for the slowest one. This internal synchronization creates significant idle time, wastes plane-level parallel resources, and further delays the request.

Motivated by the above observations, we propose a latency-consistency-aware sub-request allocation scheme called LCAS to significantly accelerate read performance by dynamically